

Adaptive Tsallis Entropy for Detecting Non-Fully-Developed Speckle in SAR Imagery

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Abstract—Detecting departures from the fully-developed speckle in Synthetic Aperture Radar (SAR) imagery is essential for reliable interpretation in remote sensing applications. We propose a statistical test based on a non-parametric estimator of Tsallis entropy, a non-additive generalization of Shannon’s formulation that provides enhanced sensitivity to heavy-tailed data. To improve accuracy with small samples, the estimator is refined through a bootstrap correction. In addition, we integrate the test within an adaptive windowing strategy that adjusts window size locally: larger in homogeneous regions to stabilize estimation, and smaller in heterogeneous areas to preserve structural details. This combination yields an unsupervised and interpretable procedure that produces p -value maps highlighting heterogeneous areas. Experimental results with both simulated data and SAR images demonstrate the robustness of the method in detecting texture variability and distinguishing between homogeneous and heterogeneous regions.

Index Terms—Tsallis entropy, Gamma distribution, G10 distribution, heterogeneity, SAR, hypothesis test

I. INTRODUCTION

SYNTHETIC Aperture Radar (SAR) technology has become a key tool in remote sensing, offering high-resolution imaging capabilities independent of solar illumination and weather conditions, which enables continuous and reliable Earth observation [1], [2]. SAR images are widely applied in environmental monitoring, disaster assessment, agriculture [3], urban planning [4], and maritime surveillance [5], among other fields [6]. However, the effective use of SAR data is hindered by speckle, a noise-like granular interference inherent to coherent imaging systems [7]. Speckle can be described as multiplicative, non-Gaussian noise, making statistical modeling and image interpretation particularly challenging [8].

Among the statistical models used for SAR intensities, the $\mathcal{G}_f^0$ distribution is widely recognized for its ability to capture different levels of scene heterogeneity. It generalizes the Gamma distribution, which represents fully developed speckle in homogeneous regions [9], [10]. While these models are effective, parameter estimation becomes unstable when applied to small local windows, which are often required to preserve spatial resolution. This limitation motivates alternative methodologies that avoid heavy reliance on parametric estimation.

Entropy has long been used as a measure of uncertainty in random phenomena, with Shannon’s entropy [11] being the

most classical form. Recent studies have shown the value of entropy-based methods in tasks such as edge detection [12], segmentation [13], and despeckling [14], motivating their extension to hypothesis testing in SAR imagery. In this realm, Frery et al. [15] and Alpala et al. [16] quantified departures from the fully-developed speckle hypothesis with the Shannon and Rényi entropy, respectively.

The Tsallis entropy [17] offers a non-additive generalization that introduces a parameter λ , allowing greater flexibility in capturing deviations from homogeneity and better sensitivity to heavy-tailed data. This makes it especially appealing for SAR analysis, where both homogeneous and highly heterogeneous areas coexist.

In this work, we propose a statistical test based on a non-parametric estimator of Tsallis entropy to distinguish between homogeneous and heterogeneous regions in SAR data. The test compares the estimated Tsallis entropy with its theoretical value under the Gamma model assumption. To improve performance in practice, we incorporate a bootstrap procedure that reduces bias in small samples. In addition, we embed the test within an adaptive windowing framework, inspired by the method of Park et al. [18]. Instead of relying on size-fixed sliding windows (e.g., 7×7), the adaptive procedure adjusts the window size locally: it enlarges in homogeneous regions to improve statistical stability, and shrinks in heterogeneous regions to avoid mixing different structures and to preserve local details.

The proposed method is unsupervised, ~~does not require training data~~, and produces interpretable p -value maps, making it suitable for scenarios where ground-truth information is scarce or unavailable.

The remainder of this paper is organized as follows. Section II reviews the statistical models for SAR data and introduces Tsallis entropy. Section III describes the proposed test and the adaptive windowing strategy. Section IV reports experiments with simulated and real SAR images. Section V concludes the paper.

II. PRELIMINARIES

A. Statistical Modeling of Intensity SAR Data

Statistical modeling plays a central role in the analysis of SAR imagery, where speckle is an inherent feature. For SAR intensity data, two main probability models are employed: the SAR-Gamma distribution (Γ_{SAR}), a reparametrization of the classical Gamma law suited to fully developed speckle, and the $\mathcal{G}_f^0$ distribution, which can describe varying levels of heterogeneity [9].

We denote $Z \sim \Gamma_{\text{SAR}}(\mu, L)$ and $Z \sim \mathcal{G}_f^0(\alpha, \gamma, L)$ when the random variable Z follows the respective distributions,

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This research, including its outputs, e.g., this manuscript, was executed in Quarto and is fully reproducible. We used RStudio version 2024.12.1+563, and R version 4.4.2. Data and code are available at: <https://github.com/rjaneth/renyi-entropy>

characterized by the following probability density functions (pdfs):

$$f_{\Gamma_{\text{SAR}}}(z; \mu, L) = \frac{L^L}{\Gamma(L) \mu^L} z^{L-1} \exp\left(-\frac{Lz}{\mu}\right), \mathbb{1}_{\mathbb{R}_+}(z) \quad (1)$$

and

$$f_{\mathcal{G}_I^0}(z; \alpha, \gamma, L) = \frac{L^L \Gamma(L - \alpha)}{\gamma^\alpha \Gamma(-\alpha) \Gamma(L)} \frac{z^{L-1}}{(\gamma + Lz)^{L-\alpha}} \mathbb{1}_{\mathbb{R}_+}(z), \quad (2)$$

where $\mu > 0$ is the mean intensity, $\gamma > 0$ is a scale parameter, $\alpha < 0$ controls the degree of texture (roughness), $L \geq 1$ is the number of looks (either nominal or estimated, thus not restricted to integer values), $\Gamma(\cdot)$ is the gamma function, and $\mathbb{1}_A(\cdot)$ is the indicator function of the set A .

The r th order moments of the $\mathcal{G}_I^0$ model are

$$E(Z^r) = \left(\frac{\gamma}{L}\right)^r \frac{\Gamma(-\alpha - r) \Gamma(L + r)}{\Gamma(-\alpha) \Gamma(L)}, \quad (3)$$

provided that $\alpha < -r$, and infinite otherwise. Therefore, assuming $\alpha < -1$, its expected value is

$$\mu = \left(\frac{\gamma}{L}\right) \frac{\Gamma(-\alpha - 1) \Gamma(L + 1)}{\Gamma(-\alpha) \Gamma(L)} = -\frac{\gamma}{\alpha + 1}. \quad (4)$$

Although the $\mathcal{G}_I^0$ distribution is defined by the parameters α and γ , in the SAR literature [19] the texture α and the mean μ are usually used. Reparameterizing (2) with μ , and denoting this model as $Z \sim \mathcal{G}_I^0(\alpha, \mu, L)$ we obtain:

$$f_{\mathcal{G}_I^0}(z; \mu, \alpha, L) = \frac{L^L \Gamma(L - \alpha)}{[-\mu(\alpha + 1)]^\alpha \Gamma(-\alpha) \Gamma(L)} \frac{z^{L-1}}{[-\mu(\alpha + 1) + Lz]^{L-\alpha}} \mathbb{1}_{\mathbb{R}_+}(z). \quad (5)$$

The Γ_{SAR} model is a particular case of the $\mathcal{G}_I^0$ distribution, as demonstrated in [9]. Specifically, for a given μ fixed,

$$f_{\mathcal{G}_I^0}(z; \mu, \alpha, L) \longrightarrow f_{\Gamma_{\text{SAR}}}(z; \mu, L) \quad \text{when } \alpha \rightarrow -\infty.$$

B. Tsallis Entropy

Tsallis entropy was originally proposed by Havrda and Charvát [20] in the context of information theory, and later extended by Tsallis [17] to emphasize its non-extensive features in statistical physics.

For a continuous random variable Z with pdf $f(z)$, the Tsallis entropy of order $\lambda \in \mathbb{R}_+ \setminus \{1\}$ is defined as:

$$T_\lambda(Z) = \frac{1}{\lambda - 1} \left[1 - \int_{\mathcal{B}} (f(z))^\lambda dz \right], \quad (6)$$

where $\mathcal{B} \subseteq \mathbb{R}$ denotes the support of $f(z)$. In the limit $\lambda \rightarrow 1$, we obtain $T_\lambda(Z) \rightarrow H(Z)$, making Tsallis a one-parameter generalization of Shannon entropy.

Using (6), we derive closed-form expressions for the Tsallis entropy of the Γ_{SAR} and the $\mathcal{G}_I^0$ distributions:

$$T_\lambda(\Gamma_{\text{SAR}}(\mu, L)) = \frac{1}{\lambda - 1} \left\{ 1 - \exp[(1 - \lambda) \ln \mu + (\lambda - 1) \ln L + \ln \Gamma(\lambda(L - 1) + 1) - \lambda \ln \Gamma(L) - (\lambda(L - 1) + 1) \ln \lambda] \right\}, \quad (7)$$

and

$$T_\lambda(\mathcal{G}_I^0(\mu, \alpha, L)) = T_\lambda(\Gamma_{\text{SAR}}(\mu, L)) + \frac{1}{\lambda - 1} \exp[(1 - \lambda) \ln \mu + (\lambda - 1) \ln L + \ln \Gamma(\lambda(L - 1) + 1) - \lambda \ln \Gamma(L)] \left\{ 1 - \exp[(1 - \lambda) \ln(-\alpha - 1) + \lambda \ln \Gamma(L - \alpha) - \lambda \ln \Gamma(-\alpha) + \ln \Gamma(\lambda(1 - \alpha) - 1) - \ln \Gamma(\lambda(L - \alpha))] \right\}. \quad (8)$$

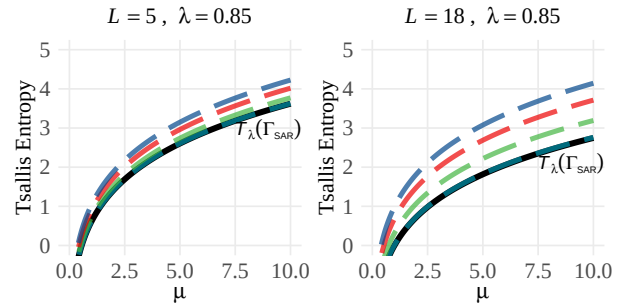
Equation (8) can be interpreted as:

$$T_\lambda(\mathcal{G}_I^0) = \underbrace{T_\lambda(\Gamma_{\text{SAR}})}_{\text{baseline entropy}} + \underbrace{\Delta_\alpha}_{\text{entropy excess caused by texture}},$$

where the extra term Δ_α captures how much entropy increases due to the texture level, caused by the roughness parameter α . When $\alpha \rightarrow -\infty$ (fully developed speckle), the excess term tends to zero, and Equation (8) reduces to the homogeneous case as in (7), i.e.,

$$\lim_{\alpha \rightarrow -\infty} T_\lambda(\mathcal{G}_I^0(\mu, \alpha, L)) = T_\lambda(\Gamma_{\text{SAR}}(\mu, L)).$$

Figure 1 illustrates the convergence of the Tsallis entropy for different values of α and $L \in \{5, 18\}$. In every case, the dashed curves of $T_\lambda(\mathcal{G}_I^0)$ approach the solid black curve of $T_\lambda(\Gamma_{\text{SAR}})$, confirming the limiting behavior.



Roughness — $\alpha = -1000$ — $\alpha = -20$ — $\alpha = -6$ — $\alpha = -2$
Figure 1: Tsallis entropy $T_\lambda(\mathcal{G}_I^0)$ converges to $T_\lambda(\Gamma_{\text{SAR}})$ as α decreases.

For the highly heterogeneous case ($\alpha = -2$), entropy values are very similar for $L = 5$ and $L = 18$, since texture dominates over speckle. In contrast, in the homogeneous limit ($\alpha \rightarrow -\infty$), texture effects vanish and entropy is mainly determined by speckle. With a low number of looks, speckle is strong and increases the uncertainty, which raises the entropy values. However, the curves for different α values appear more concentrated, since the contribution of texture to entropy is masked by the dominant speckle. As the number of looks increases, speckle is progressively reduced and entropy decreases. This reduction of speckle makes the entropy excess Δ_α more evident, resulting in more separated curves across α .

C. Non-parametric Estimation of Tsallis Entropy

Following the spacing-based approach of Vasicek [21] and Ebrahimi et al. [22] for Shannon entropy, and the methodology

for Rényi entropy proposed by Al-Labadi et al. [23], we derive a nonparametric estimator for Tsallis entropy.

Let $Q(p) = F^{-1}(p)$ denote the quantile function for $p \in (0, 1)$. From (6), the inner integral can be expressed as an expectation under a random variable Z with pdf f :

$$\int_{\mathcal{B}} f(z)^\lambda dz = \int_{\mathcal{B}} f(z)^{\lambda-1} f(z) dz = \mathbb{E}[f(Z)^{\lambda-1}].$$

Using the change of variables $p = F(z)$ (hence $dp = f(z) dz$) and the identity $f(Q(p)) = 1/Q'(p)$, we obtain

$$\mathbb{E}[f(Z)^{\lambda-1}] = \int_0^1 (Q'(p))^{1-\lambda} dp,$$

which yields the representation

$$T_\lambda(Z) = \frac{1}{\lambda-1} \left\{ 1 - \int_0^1 (Q'(p))^{1-\lambda} dp \right\}. \quad (9)$$

Let $\mathbf{Z} = (Z_1, Z_2, \dots, Z_n)$ be an i.i.d. sample from F , with order statistics $Z_{(1)} \leq Z_{(2)} \leq \dots \leq Z_{(n)}$. For a window size $m \in \{1, \dots, \lfloor (n-1)/2 \rfloor\}$, define the m -spacing

$$D_{i,m} := Z_{(i+m)} - Z_{(i-m)}, \quad i = 1, 2, \dots, n,$$

with boundary conventions $Z_{(i-m)} := Z_{(1)}$ if $i \leq m$ and $Z_{(i+m)} := Z_{(n)}$ if $i \geq n-m$, where $\lfloor \cdot \rfloor$ denotes the floor function. The m -spacing density estimator at $Z_{(i)}$ is

$$\hat{f}_n(Z_{(i)}) = \frac{(m/n) c_i}{Z_{(i+m)} - Z_{(i-m)}}, \quad i = 1, 2, \dots, n, \quad (10)$$

where

$$c_i = \begin{cases} \frac{m+i-1}{m} & \text{if } 1 \leq i \leq m, \\ 2 & \text{if } m+1 \leq i \leq n-m, \\ \frac{n+m-i}{m} & \text{if } n-m+1 \leq i \leq n. \end{cases}$$

By construction, $\hat{f}_n(Z_{(i)}) \approx f(Z_{(i)})$. Since $f(Q(p)) Q'(p) = 1$, taking the reciprocal yields an estimator of the quantile derivative:

$$\widehat{Q}'(p_i) = \frac{1}{\hat{f}_n(Z_{(i)})} = \frac{D_{i,m}}{(m/n) c_i}, \quad p_i \in (0, 1). \quad (11)$$

Approximating the integral in (9) by a Riemann sum over the grid $p_i = i/n$ and inserting $\widehat{Q}'(p_i)$ from (11) yields the nonparametric Tsallis entropy estimator:

$$\begin{aligned} \hat{T}_\lambda(\mathbf{Z}) &= \frac{1}{\lambda-1} \left\{ 1 - \frac{1}{n} \sum_{i=1}^n [\widehat{Q}'(p_i)]^{1-\lambda} \right\} \\ &= \frac{1}{\lambda-1} \left\{ 1 - \frac{1}{n} \sum_{i=1}^n \left(\frac{Z_{(i+m)} - Z_{(i-m)}}{(m/n) c_i} \right)^{1-\lambda} \right\}. \end{aligned} \quad (12)$$

As $\lambda \rightarrow 1$, T_λ converges to Shannon entropy and $\hat{T}_\lambda$ recovers the Vasicek estimator [21]. This estimator is asymptotically consistent, i.e., it converges in probability to the true value T_λ when $m, n \rightarrow \infty$ and $m/n \rightarrow 0$. We use the heuristic spacing $m = \lfloor \sqrt{n} + 0.5 \rfloor$.

III. PROPOSED METHODOLOGY

A. Optimal λ value

We investigate the optimal order $\sim \lambda$ for the Tsallis entropy estimator using simulated samples of size $n = 49$ drawn from $Z \sim \Gamma_{\text{SAR}}(1, 5)$. The bias and mean squared error (MSE) were evaluated for varying values of λ through a Monte Carlo experiment.

As shown in Fig. 2, for $L > 1$ we found that $\lambda = 0.85$ yields the lowest MSE while maintaining a low bias, thus providing a favorable trade-off between bias and variance.

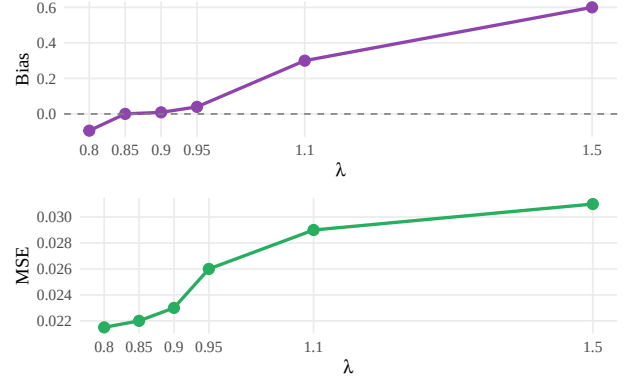


Figure 2: Bias and MSE as a function of λ , for the Tsallis entropy estimator, with $n = 49$ and $L = 5$.

B. Bootstrap Correction for Entropy Estimator

Following Refs. [15], [16], [24], we refine the non-parametric entropy estimator $\hat{T}_\lambda$ defined in (12) with bootstrap, obtaining:

$$\tilde{T}_\lambda = 2\hat{T}_\lambda(\mathbf{Z}) - \frac{1}{B} \sum_{b=1}^B \hat{T}_\lambda(\mathbf{Z}^{(b)}),$$

where B is the number of bootstrap replications, and $\mathbf{Z}^{(b)}$ denotes the b -th resampled dataset obtained by drawing n observations with replacement from $\mathbf{Z}$.

A Monte Carlo study with 1000 replications for each sample size $n \in \{9, 25, 49, 81, 121\}$ from the $\Gamma_{\text{SAR}}(1, 5)$ confirms that for $\lambda = 0.85$, the bootstrap-corrected estimator $\tilde{T}_\lambda$ with $B = 200$, reduces both bias and MSE compared to the original estimator $\hat{T}_\lambda$, with significant improvements for small sample sizes, as shown in Fig. 3.

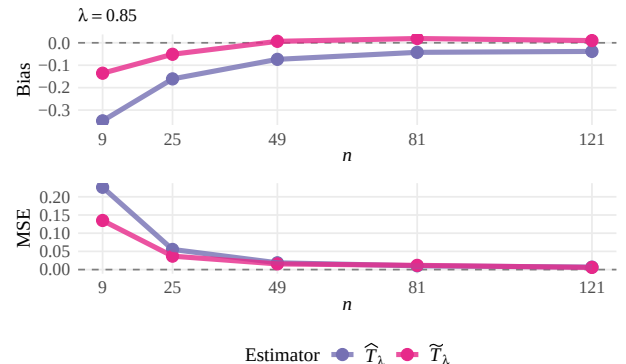


Figure 3: Bias and MSE for the Tsallis entropy estimators of Γ_{SAR} , with $L = 5$.

From this point forward, all subsequent simulations and comparisons will be based on the improved bootstrap estimator $\tilde{T}_\lambda$.

C. Hypothesis Testing

We test whether the observed data come from a homogeneous (Γ_{SAR}) or a heterogeneous ($\mathcal{G}_I^0$) region, as follows:

$$\begin{cases} \mathcal{H}_0 : \mathbb{E}[\tilde{T}_\lambda] = T_\lambda(\Gamma_{\text{SAR}}) & (\text{Homogeneous region}), \\ \mathcal{H}_1 : \mathbb{E}[\tilde{T}_\lambda] = T_\lambda(\mathcal{G}_I^0) & (\text{Heterogeneous region}). \end{cases} \quad (13)$$

Under $\mathcal{H}_0$, the expected value of the entropy estimator should match the theoretical $T_\lambda(\Gamma_{\text{SAR}})$. Meanwhile, significant deviations indicate heterogeneity.

D. The Proposed Test

We propose a **statistic** test that identifies the discrepancy between estimated and theoretical entropy under homogeneity. Since the number of looks $L \geq 1$ is known, we define the test **statistic** as follows:

$$S_{\tilde{T}_\lambda}(\mathbf{Z}; L) = \tilde{T}_\lambda - \left\{ \frac{1}{\lambda - 1} \left[1 - \exp\left((1 - \lambda) \ln \hat{\mu} + (\lambda - 1) \ln L + \ln \Gamma(\lambda(L - 1) + 1) - \lambda \ln \Gamma(L) - (\lambda(L - 1) + 1) \ln \lambda\right) \right] \right\}, \quad (14)$$

where $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n Z_i$ is the sample mean.

This **statistic** can be interpreted as:

$$S_{\tilde{T}_\lambda} = \underbrace{\tilde{T}_\lambda}_{\text{estimated}} - \underbrace{T_\lambda(\Gamma_{\text{SAR}})}_{\text{theoretical under } H_0},$$

the difference between the estimated entropy and the expected value under homogeneity. Values close to zero indicate that the region behaves like fully developed speckle, while large positive values signal excess entropy and, thus, heterogeneity. This formulation avoids the need to estimate $\mathcal{G}_I^0$ parameters such as α , offering a simple, interpretable, and statistically grounded test. Moreover, the method remains effective for small samples, aided by the bootstrap bias correction.

Fig. 4 shows the empirical distribution of $S_{\tilde{T}_\lambda}$ under $\mathcal{H}_0$, obtained from 10^4 Monte Carlo simulations with **varying** sample sizes ($n \in \{49, 81, 121\}$), $\lambda = 0.85$, and $L \in \{5, 18\}$. The empirical densities are tightly concentrated around zero, confirming that under $\mathcal{H}_0$ the test **statistic** has mean approximately 0; at the same time, their moderately heavy tails reveal sensitivity to departures from homogeneity, which is desirable for detecting subtle texture.

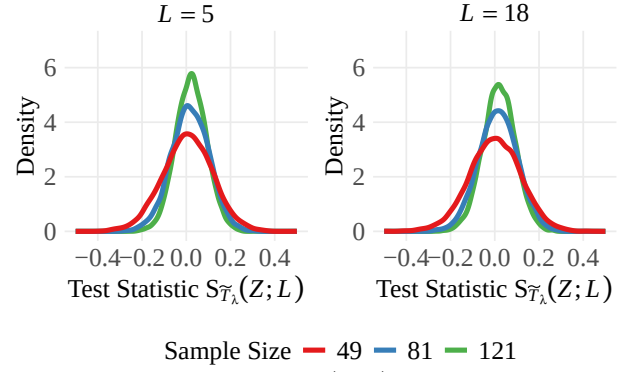


Figure 4: Empirical densities of $S_{\tilde{T}_\lambda}(\mathbf{Z}; L)$ under $\mathcal{H}_0$, with $\lambda = 0.85$.

Under the asymptotic properties of the entropy estimators [21], [25], for sufficiently large samples, $S_{\tilde{T}}(\mathbf{Z}; L)$ follows an asymptotic normal distribution:

$$S_{\tilde{T}}(\mathbf{Z}; L) \xrightarrow[n \rightarrow \infty]{\mathcal{D}} \mathcal{N}(\mu_S, \sigma_S^2),$$

where $\xrightarrow{\mathcal{D}}$ denotes convergence in distribution.

Here, $\mu_S = \mathbb{E}[S_{\tilde{T}}(\mathbf{Z}; L)]$ and $\sigma_S^2 = \text{Var}[S_{\tilde{T}}(\mathbf{Z}; L)]$ are the theoretical mean and variance of the test statistic under $\mathcal{H}_0$. This asymptotic normality can be explained by noting that the entropy estimator (and hence $S_{\tilde{T}}$) can be expressed as a sum or average of numerous observations; therefore, by the Central Limit Theorem and the delta method, its sampling distribution approaches a Gaussian form for large n .

In practice, we estimate $\hat{\mu}_S$ and $\hat{\sigma}_S$ via Monte Carlo under $\mathcal{H}_0$.

$$\varepsilon = \frac{S_{\tilde{T}}(\mathbf{Z}; L) - \hat{\mu}_S}{\hat{\sigma}_S},$$

which is asymptotically standard normal distributed for large n . Consequently, two-sided p -values are obtained as $2\Phi(-|\varepsilon|)$, where $\Phi(\cdot)$ is the cumulative distribution function of the standard normal distribution.

The practical procedure is as follows: first, compute the test statistic $S_{\tilde{T}}(\mathbf{Z}; L)$ for the observed data; then standardize it using the estimated mean and standard deviation ($\hat{\mu}_S, \hat{\sigma}_S$); next, compute the p -value as $2\Phi(-|\varepsilon|)$; and finally, make a decision by comparing the p -value with the threshold 0.05, as illustrated in Fig. 5.

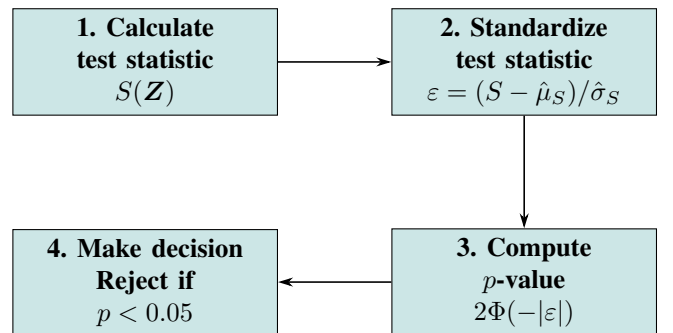


Figure 5: Workflow of the entropy-based hypothesis test.

E. Size and Power Analysis of the Proposed Test

The statistical validity and effectiveness of the proposed Tsallis entropy-based test was examined through two funda-

mental properties: *size* and *power*. These provide insight into the probability of incorrect decisions in hypothesis testing.

The size of a statistical test, also known as the Type I error rate, refers to the probability of incorrectly rejecting the null hypothesis $\mathcal{H}_0$ when it is in fact true. In hypothesis testing, practitioners typically specify a nominal significance level, commonly set at 1 %, 5 %, and 10 %, which represent acceptable probabilities of committing a Type I error. A well-calibrated test should reproduce these levels empirically across sample sizes and conditions.

To assess this, we performed 1000 Monte Carlo replications under $\mathcal{H}_0$ with data from a Γ_{SAR} distribution ($\mu = 1$), computing the test statistic via a bootstrap entropy estimator ($B = 100$, $\lambda = 0.85$). The empirical size was then estimated as the proportion of replications in which the null hypothesis was incorrectly rejected. The observed Type I error rates remained close to the nominal levels, confirming the validity and proper calibration of the proposed procedures.

The power of a test is its probability of correctly rejecting $\mathcal{H}_0$ when it is false. Equivalently, it equals $1 - \beta$, where β is the Type II error rate. High power thus indicates strong sensitivity to departures from $\mathcal{H}_0$. To assess power, we simulated data under the alternative hypothesis $\mathcal{H}_1$, assuming a $\mathcal{G}_I^0$ distribution with $\mu = 1$ and $\alpha = -2$. Again, 1000 Monte Carlo replications were performed for each configuration of sample size and L , and the test statistic was computed using the bootstrap estimator ($B = 100$, $\lambda = 0.85$).

As expected, power increased with both the sample size and the number of looks, confirming the effectiveness of the Tsallis entropy-based test in detecting departures from the null hypothesis.

The complete results for both size and power are reported in Table I.

TABLE I
SIZE AND POWER OF THE $S_{\tilde{\tau}_\lambda}(\mathbf{Z})$ TEST STATISTIC (TSALLIS).

L	n	Size			Power		
		1 %	5 %	10 %	1 %	5 %	10 %
5	25	0.014	0.051	0.103	0.958	0.991	0.990
	49	0.011	0.048	0.107	0.990	1.000	0.999
	81	0.012	0.057	0.101	0.997	0.998	0.999
	121	0.014	0.060	0.116	0.999	0.999	0.997
8	25	0.010	0.049	0.108	0.992	1.000	1.000
	49	0.008	0.055	0.108	1.000	0.999	1.000
	81	0.012	0.052	0.097	0.998	1.000	0.999
	121	0.016	0.072	0.116	0.998	1.000	0.999
18	25	0.015	0.052	0.111	1.000	1.000	1.000
	49	0.013	0.046	0.099	1.000	1.000	1.000
	81	0.013	0.047	0.105	1.000	1.000	1.000
	121	0.012	0.065	0.109	1.000	1.000	1.000

F. Adaptive Windows with Tsallis Entropy

Our approach builds on the adaptive windowing algorithm for SAR images proposed by Park et al. [18], which adjusts the analysis window size according to the local homogeneity of the scene. For each pixel (i, j) , the coefficient of variation is computed over the border of the current window,

$$C_{ij} = \frac{\sigma_{ij}}{\mu_{ij}}, \quad (15)$$

where μ_{ij} and σ_{ij} are calculated using only the border pixels b_{ij} of the window w_{ij} . In homogeneous areas, $C_{ij} \approx \sigma_n$, with

$$\sigma_n = \frac{0.523}{\sqrt{L}}. \quad (16)$$

The homogeneity criterion is defined by comparing C_{ij} with an adaptive threshold U_{ij} , given by

$$U_{ij} = \eta \left(1 + \sqrt{\frac{1 + 2\sigma_n^2}{8(W_{ij} - 1)}} \right) \sigma_n, \quad (17)$$

where $W_{ij} = 2N_{ij} + 1$ is the current window side length (with radius N_{ij}), and η is a tuning parameter controlling the degree of smoothing. Values close to $\eta = 1.0$ provide limited speckle suppression, while larger values increase smoothing and yield stronger noise reduction.

The window radius N_{ij} is updated dynamically and locally as follows:

$$N_{i,j+1} = \begin{cases} \min(N_{ij} + 1, N_{\max}), & \text{if } C_{ij} \leq U_{ij}, \\ \max(N_{ij} - 1, N_{\min}), & \text{if } C_{ij} > U_{ij}, \end{cases}$$

here, $N_{\max}$ and $N_{\min}$ denote the maximum and minimum window size, respectively.

The procedure can be summarized as:

- **Homogeneous regions** ($C_{ij} \leq U_{ij}$): the window grows, up to $N_{\max}$, which enhances speckle reduction and stabilizes the estimates.
- **Heterogeneous regions** ($C_{ij} > U_{ij}$): the window shrinks, down to $N_{\min}$, which preserves edges and prevents the mixing of different classes.

Once the window radius N_{ij} has been determined for pixel (i, j) , the corresponding full window is extracted and the proposed test in (14) is applied to obtain the local statistic. This produces two outputs:

- a map of test statistics over the image, and
- a window-size map W_{ij} , which records the locally selected window side (larger in homogeneous regions, smaller in heterogeneous ones).

For inference, each value of the test statistic is standardized using the empirical mean and standard deviation of the image, and two-sided p -values are then computed.

This adaptive strategy has several advantages. It is locally responsive to image texture, using small windows in heterogeneous areas and larger ones in homogeneous zones. This ensures edge preservation and statistical stability, while the parameter η controls the balance between smoothness and detail preservation. Moreover, the framework is extensible and can be applied with any entropy estimator.

IV. RESULTS

A. Analysis with Simulated Data

Figure 6(a) presents the structural design of the synthetic phantom with dimensions of 500×500 pixels, proposed by Gomez et al. [26] as a tool to assess the performance of

speckle-reduction filters. This layout serves as a base template to generate the simulated images shown in Figures 6(b).

Each region is filled with data simulated from the G_I^0 distribution (5), using different combinations of the roughness parameter α and the mean μ , as indicated by the labels in each quadrant of the image. Light regions correspond to textured observations (heterogeneous), while darker regions represent textureless areas (homogeneous).

The α parameter of the G_I^0 distribution is essential for interpreting texture characteristics. Values ~~near zero greater than -3~~ suggest extremely textured targets, such as urban zones [27]. As the α value decreases, it indicates regions with moderate texture (in the $[-6, -3]$ region), related to forest zones, while values below -6 correspond to textureless regions, such as pasture, agricultural fields, and water bodies [28].

The proposed Tsallis-based test statistic, S_{T_λ} , was applied to the simulated image using the adaptive windowing strategy with ~~side lengths~~ ranging from 5×5 (minimum) to 11×11 (maximum), with the smoothing parameter set to $\eta = 5$. Figure 6(c) displays the resulting p -value map for $L = 9$.

The color gradient provides an interpretable scale: dark red regions ($p \approx 0$) indicate strong evidence of heterogeneity, consistently appearing in highly textured zones such as those with $\alpha > -3$, while dark blue regions ($p \approx 1$) correspond to homogeneous areas, typically observed for very low α values (e.g., $\alpha < -6$). Intermediate colors, such as yellow and green ($p \approx 0.2$ – 0.6), represent transitional responses that capture moderately textured areas (e.g., $-6 \leq \alpha \leq -3$), whereas light blue tones ($p \approx 0.7$ – 0.9) occur in regions that are close to homogeneous but still contain subtle texture variations. This ~~gradation of colors~~ highlights the sensitivity of the proposed test: instead of producing ~~only~~ binary outcomes, it provides a continuous measure of heterogeneity across the image, allowing areas with intermediate texture levels to be represented by intermediate p -values that reflect the gradual transition between homogeneous and heterogeneous scattering conditions.

B. Applications to SAR Data

The selected scenes, acquired in HH polarization, correspond to the outskirts of Munich and Dublin Port (Ireland), as shown in Figs. 7(a)–8(a). Corresponding optical images (Figs. 7(b)–8(b)) ~~illustrate~~ land cover context.

Table II details the SAR acquisition parameters.

TABLE II
PARAMETERS OF SELECTED SAR IMAGES

Scene	Mission	Band	Size (pixels)	L	Resolution [m]	Acquisition Date
Munich	UAVSAR	L	1024×1024	12	4.9×7.2	16-04-2015
Dublin	TanDEM-X	X	1100×1100	16	1.35×1.35	03-09-2017

We applied the proposed test statistic S_{T_λ} to real SAR data using adaptive local sliding windows ranging from 5×5 to 11×11 , with the smoothing parameter set to $\eta = 5$. The results are illustrated in Figures 7(c)–8(c), which display the original SAR image, the corresponding optical reference, and the resulting p -value maps.

The p -value maps ~~provide~~ a clear separation between homogeneous and heterogeneous areas. Dark-red tones ($p < 0.05$) cluster around highly textured features, including urban blocks, industrial and harbor facilities, and road networks. These regions exhibit strong departures from homogeneity, consistent with the expected scattering variability in built environments. Intermediate tones (green and yellow, $0.2 < p < 0.6$) appear in semi-structured areas such as agricultural fields and mixed land covers, reflecting moderate texture. By contrast, blue tones ($p \approx 1$) dominate in smooth regions such as open water bodies, uniform pastures, and large homogeneous crop zones, where the data behave in accordance with the null hypothesis.

The adaptive nature of the windowing plays a key role in these results. Larger windows are automatically selected in homogeneous regions, enhancing stability and noise suppression, while smaller windows are used in heterogeneous areas, ensuring edge preservation and avoiding class mixing. This balance allows the method to capture fine-scale details such as canals, field boundaries, and narrow urban structures, while still providing robust detection of broader homogeneous areas.

V. CONCLUSIONS

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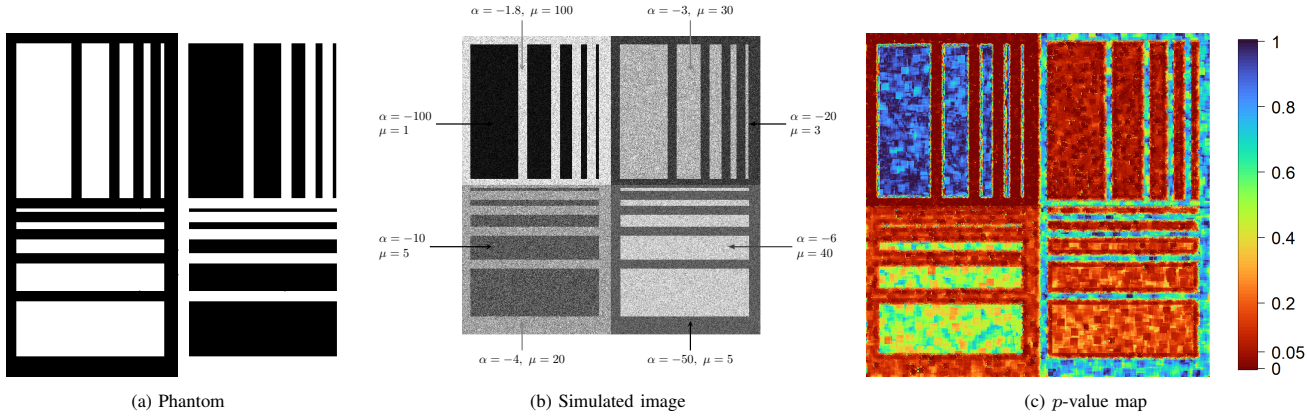


Figure 6: Results on simulated data using the test statistic $S_{\bar{T}_\lambda}$ with adaptive windows ranging from 5×5 to 11×11 . The image has size 500×500 pixels, with $L = 9$.

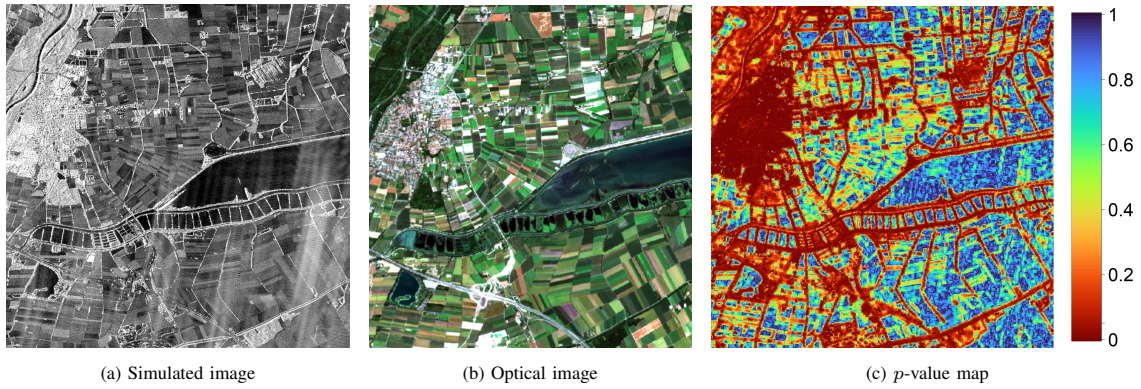


Figure 7: Results on SAR data-Munich with the test $S_{\bar{T}_\lambda}$ and adaptive windows (5×5 to 11×11).

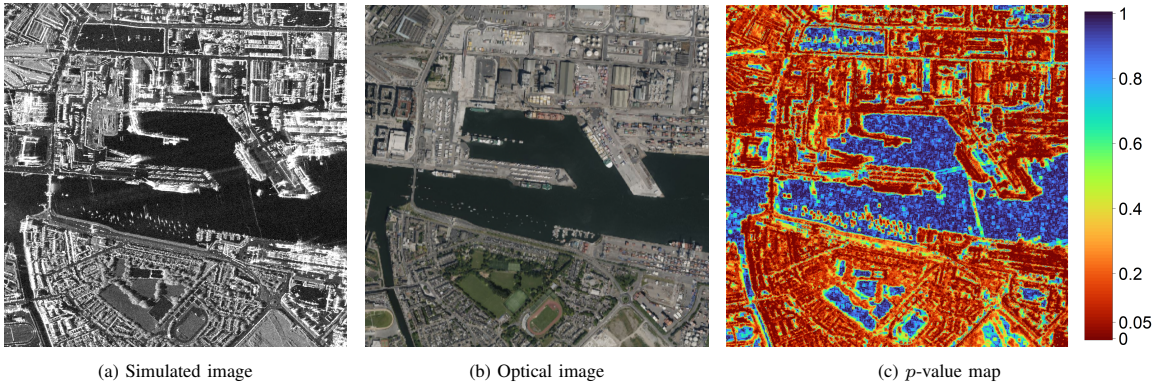


Figure 8: Results on SAR data-Dublin with the test $S_{\bar{T}_\lambda}$ and adaptive windows (5×5 to 11×11).

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